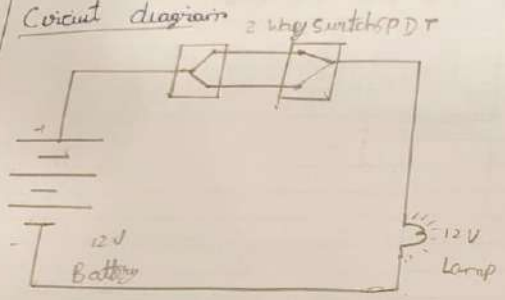
## Staircase wiring

Circuit diagram




SPDT- Single pole double Throw switch

| Mode | $S_1$ | $S_2$ | Lamp |
|---|---|---|---|
| 1 | ON | ON | ON |
| 2 | ON | OFF | OFF |
| 3 | OFF | ON | ON |
| 4 | OFF | OFF | OFF |

| Components | Proteus | Specification |
|---|---|---|
| AC supply | Vsine | Amplitude 230 V |
| Two way Switch | SPDT | Frequency 50 Hz |
| Lamp | Anunsated | 230 V |

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## STAIRCASE WIRING

Aim:-
To develop and test the staircase wiring using Proteus

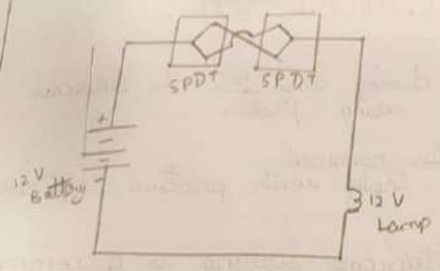
Apparatus required:-
Laptop with proteus software

Theory:-
Staircase wiring is a common multiple way switching or two way switching connection. It involves wiring for one light with two switches. In this set up is controlled by two switches located operate the load from separate position above or below the staircase from inside or outside of room or two way lod switch.

Procedure:-

- Drag the required components from the process library
- Connect the components as per circuit diagram.
- Run the simulation and check the lamp for different switching condition

# Cross Connection




| Mode | $S_1$ | $S_2$ | Lamp |
|---|---|---|---|
| 1 | ON | ON | OFF |
| 2 | ON | OFF | ON |
| 3 | OFF | ON | ON |
| 4 | OFF | OFF | OFF |

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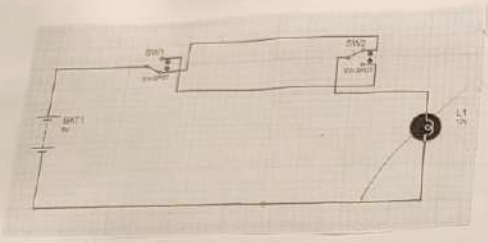


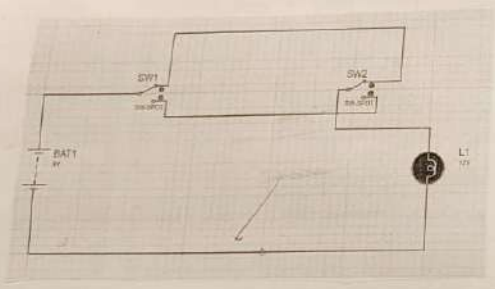


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Result:-

Thus the staircase using connection wire developed and tested successfully